DO NOW — RECALL

Lesson 114 · Compilers, Interpreters and Assemblers

Name: _____ Class: _____ Date: _____

1. Does free-of-charge imply that software is open source?

2. A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

3. For the binary value 1010, what is its decimal value?

4. A linked-list node has data and what other essential field?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

No. Price is not the same as licence/source rights.

2. Model answer

Sum 0 and carry-out 1.

3. Model answer

10.

4. Model answer

A link/reference to the next node (for a singly linked list).

By the end, you can…

• Translates a complete source unit to target code before execution, often enabling optimisation.

• Executes by translating/evaluating during running; behaviour varies by implementation.

• Converts symbolic assembly mnemonics into machine instructions.

Vocabulary — know these terms

compiler — Translates source to target code before execution.

interpreter — Executes by translating/evaluating at runtime.

assembler — Turns assembly into machine/object code.

object code — Compiled code awaiting linking or execution.

NEW LEARNING

Read, annotate and explain

Compiler

Translates a complete source unit to target code before execution, often enabling optimisation.

Interpreter

Executes by translating/evaluating during running; behaviour varies by implementation.

Assembler

Converts symbolic assembly mnemonics into machine instructions.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A team distributes an executable without its original C source. Explain the compiler's role.

Model answer

A compiler translates source to target machine/object code; linked executable can run without the source being present. Compilation can report syntax errors and apply optimisation before execution.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Distinguish compiler, interpreter and assembler with a relevant input for each. [6]

Question 2

Explain two advantages of compiling a program before distribution. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A Python-like runtime, C compiler and LMC assembler are used in one project. Compare debugging, portability and execution trade-offs. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An assembler translates high-level Python syntax into binary machine code.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.